

## 07 | Beyond 180°: restoration, trajectory and 3D accuracy

### 07 | 超過 180°：影像還原、軌跡與三維精度

Research note / 研究札記 · 2026-09-29 · English + 繁體中文

#### 1. There is evidence, but not one interchangeable accuracy claim / 有證據，但各種精度不能互換

EN. Literature reports calibration of  $>180^\circ$  lenses, use of rearward rays in motion estimation, and native rendering of  $200^\circ$  fisheye images. These establish different capabilities. “Image restoration” may mean geometric remapping, novel-view rendering, or recovering a metric scene; each needs a different test. To substantiate rear-field 3D metrology, a study must identify observations with  $\theta > 90^\circ$ , use independent metric references, and report error specifically for that region. A nominal lens FoV or a whole-image average does not satisfy all three conditions.

中。文獻報告了超過  $180^\circ$  鏡頭的校正、後向光線用於運動估計，以及  $200^\circ$  魚眼影像的原生渲染；它們證明的是不同能力。「影像還原」可能指幾何重映射、新視角渲染，或恢復公制場景，各自需要不同測試。要支持後半球三維度量，研究必須指出  $\theta > 90^\circ$  的觀測、使用獨立公制基準，並單獨報告該區域誤差。標稱鏡頭 FoV 或全影像平均，不能同時滿足這三項條件。

#### 2. Calibration evidence is not a rear-field metric benchmark / 校正證據不等於後半球公制基準

EN. The original KB paper's  $190^\circ$  experiment demonstrates that KB is not inherently limited to  $180^\circ$ . Its validation is in reprojection pixels, not object length. Likewise, the  $185^\circ$  wand experiment in topic 04 measures physical lengths but does not isolate which measurements are supported by rearward pixels. These are useful complementary results, yet neither supplies a  $\theta > 90^\circ$ -only coordinate-error distribution. Full FoV= $190^\circ$  reaches  $\theta_{\max}=95^\circ$ , which is also much less demanding in angular coverage than a proposed  $280^\circ$  lens reaching  $\theta_{\max}=140^\circ$ . [KB06, WAND14]

中。KB 原始論文的  $190^\circ$  實驗證明 KB 本身並不受限於  $180^\circ$ ，但其驗證單位是重投影像素，而不是物件長度。主題 04 的  $185^\circ$  量測桿實驗雖然測量實體長度，卻沒有分離哪些結果由後向像素支持。兩者提供互補證據，但都沒有提供僅針對  $\theta > 90^\circ$  的座標誤差分布。完整 FoV= $190^\circ$  只到  $\theta_{\max}=95^\circ$ ，其角度覆蓋要求也遠低於  $280^\circ$  鏡頭的  $\theta_{\max}=140^\circ$ 。[KB06, WAND14]

EN. Scaramuzza's ICVS 2006 experiment is another historical example: it reports vertical coverage above  $200^\circ$  for a catadioptric camera and reconstructs a known trihedron. However, it does not report the checker-size error separately for rearward angular bins. A  $>200^\circ$  camera plus an object reconstruction is therefore relevant evidence, but still does not establish a rear-field-only dimensional tolerance. [OC06]

中。Scaramuzza 的 ICVS 2006 實驗也是歷史例子：它報告折反射相機的垂直視域超過  $200^\circ$ ，並重建已知三面體；但沒有針對後向角度分組另報棋盤格尺寸誤差。因此，「超過  $200^\circ$  的相機加上物件重建」是相關證據，仍不足以建立僅針對後半球的尺寸公差。[OC06]

### 3. Explicit rearward-ray evidence: LF-VIO / 明確的後向光線證據：LF-VIO

EN. LF-VIO evaluates a panoramic annular camera covering  $360^\circ$  azimuth and polar angles  $40\text{--}120^\circ$ , including negative- $z$  directions. This is an annulus with a central blind region, not a circular  $240^\circ$  fisheye cap. Its PALVIO experiment uses motion capture and compares angular subsets. In Table II, sequence ID06 has absolute trajectory error of 0.093 m for the full tested field versus 0.124 m for the  $40\text{--}90^\circ$  subset. This supports the usefulness of rearward observations in that visual-inertial system; it does not measure the dimensional accuracy of objects seen exclusively behind the camera. [LF22, section IV-C, Table II]

中。LF-VIO 評估方位角  $360^\circ$ 、極角  $40\text{--}120^\circ$  的全景環形相機，包含負  $z$  方向。這是具有中央盲區的環形視域，不是完整  $240^\circ$  圓形魚眼球冠。其 PALVIO 實驗使用動作捕捉並比較不同角度子集。在表 II，ID06 序列使用完整測試視域的絕對軌跡誤差為 0.093 公尺，僅使用  $40\text{--}90^\circ$  子集則為 0.124 公尺。這支持後向觀測對該視覺慣性系統的效益，卻沒有量測完全位於相機後方物件的尺寸精度。[LF22，第 IV-C 節、表 II]

### 4. Native $200^\circ$ rendering evidence / 原生 $200^\circ$ 渲染證據

EN. Gunes et al.'s August 2025 v1 evaluates Fisheye-GS and 3DGUT with real  $200^\circ$  imagery and compares  $200^\circ$ ,  $160^\circ$  and  $120^\circ$  conditions. Fisheye-GS benefits from reducing FoV to  $160^\circ$ , whereas 3DGUT remains more stable at  $200^\circ$ . Evaluation uses held-out views and PSNR, SSIM and LPIPS: these are appearance metrics. They do not establish millimetre geometry, calibrated physical scale, or an independent error bound for the  $90\text{--}100^\circ$  annulus. The study provides evidence of rendering feasibility; it must not be recast as certified object-space measurement. This note cites v1 explicitly because the current arXiv record has a revised title. [GS25v1]

中。Gunes 等人的 2025 年 8 月 v1 使用真實  $200^\circ$  影像評估 Fisheye-GS 與 3DGUT，並比較  $200^\circ$ 、 $160^\circ$ 、 $120^\circ$  條件。Fisheye-GS 在縮小到  $160^\circ$  後受益，而 3DGUT 在  $200^\circ$  下較穩定。評估使用保留視角與 PSNR、SSIM、LPIPS，這些都是外觀指標，不能證明毫米級幾何、已校驗實體尺度，或  $90\text{--}100^\circ$  環帶的獨立誤差界限。該研究提供渲染可行性證據，不應被改述為經認證的物件空間度量。由於目前 arXiv 記錄已更改標題，本文明確引用 v1。[GS25v1]

### 5. FIORD: useful data, with important evaluation limits / FIORD：有用資料及其評估限制

EN. FIORD supplies indoor/outdoor fisheye imagery and laser-scanner reference data, making it relevant to future metric evaluation. However, section 3.5 of the reviewed manuscript rectifies images into a pinhole-compatible format for its baseline rendering experiments. Section 3.4 describes scale-inclusive alignment using 7-10 correspondences and gives a 25 cm alignment RMSE example for the largest indoor scene. That is an alignment statistic, not a rear-annulus reconstruction error or universal ground-truth accuracy. Before claiming fine metric performance on this dataset, quantify registration uncertainty and use independent checks. [FIORD25]

中。FIORD 提供室內外魚眼影像及雷射掃描基準，有助未來的公制度量評估。然而，本次查閱稿件第 3.5 節將影像校正為針孔相容格式，用於基準渲染實驗；第 3.4 節則利用 7-10 組對應點進行包含尺度的對齊，並對最大室內場景舉出 25 公分配準 RMSE。那是配準統計，不是後半球環帶重建誤差，也不是普遍的真值精度。若要利用此資料集主張精細公制效能，必須先量化配準不確定度，並使用獨立檢核。[FIORD25]

## 6. What has not been established here / 本次尚未證實的部分

EN. In the sources verified for this review, no study establishes a general dimensional tolerance for the  $\theta=90-140^\circ$  region of a single  $280^\circ$  lens, nor a controlled KB-versus-OCam accuracy ranking there. This is a bounded literature finding, not proof that no such paper exists. A dual-lens  $360^\circ$  camera does not automatically validate one lens over  $280^\circ$ ; nominal coverage, sensor cropping, lens obstruction and calibrated angular coverage differ. The proposal should frame rear-field metrology as a question to test rather than an already demonstrated consequence of model support.

中。在本次核對的來源中，尚未有研究建立單一  $280^\circ$  鏡頭  $\theta=90-140^\circ$  區域的通用尺寸公差，也沒有在該區域控制比較 KB 與 OCam 精度。這是有範圍限制的文獻查核結果，不是證明世上不存在相關論文。雙鏡頭  $360^\circ$  相機不會自動驗證單鏡頭的  $280^\circ$  能力；標稱覆蓋、感光元件裁切、鏡體遮蔽與已校正角度範圍都不同。研究計畫應將後半球度量視為待檢驗問題，而不是模型支援後便已證明的結果。

## 7. A concrete rear-field validation protocol / 具體的後半球驗證程序

EN. Place surveyed 3D targets and certified lengths across  $\theta$  bins  $0-60^\circ$ ,  $60-85^\circ$ ,  $85-95^\circ$ ,  $95-110^\circ$  and then up to the actual calibrated limit, with several azimuths and distances. These are suggested bins; adjust them to sample availability. Acquire stereo or multi-view overlap so rearward targets have non-degenerate intersections. Compare front-only, rear-only and mixed observations with fixed and unrestricted feature budgets. Keep validation controls out of calibration and registration. Report bearing error, independent XYZ errors, length bias/RMS/P95, completeness and failure rate per bin. Also report how much of each bin is genuinely observed rather than interpolated.

中。在  $\theta$  分組  $0-60^\circ$ 、 $60-85^\circ$ 、 $85-95^\circ$ 、 $95-110^\circ$ ，以及更外側直到實際校正極限，配置經測量的三維標靶及經校驗長度，涵蓋多個方位與距離。這些是建議分組，應依實際取樣調整。拍攝具有立體或多視角重疊的資料，確保後向標靶有非退化交會。比較前向限定、後向限定及混合觀測，並分別使用固定及不限特徵數。驗證控制資料不可參與校正或配準。逐組報告視線角度誤差、獨立 XYZ 誤差、長度偏差 / RMS / P95、完整率與失敗率，也要說明每組實際觀測覆蓋與插值部分的比例。

EN. For geometric image remapping, compare rays or independently projected fiducials, and state the output projection and sampling density. For novel-view synthesis, report angularly masked appearance scores as well as whole-image scores; when comparing FoVs, include the same common angular region. For 3D metrology, retain metric units and independent scale checks. A successful result in one evaluation track should not substitute for the other two. This protocol is a proposed research design; no new calibration or reconstruction experiment was performed for these documents.

中。幾何影像重映射應比較光線或獨立投影標記，並說明輸出投影及取樣密度。新視角合成應同時報告角度遮罩內及全影像外觀分數；比較不同 FoV 時，也應比較相同的共同角度區域。三維度量則保留公制單位及獨立尺度檢核。任何一類評估成功，都不能替代其他兩類。以上為建議研究設計；本次文件未執行新的校正或重建實驗。

## Sources / 文獻來源

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